

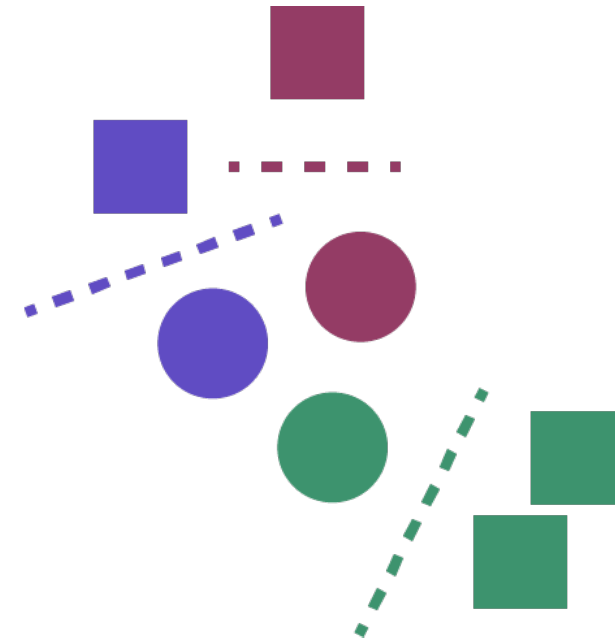
Group eXplanations with linear approximation

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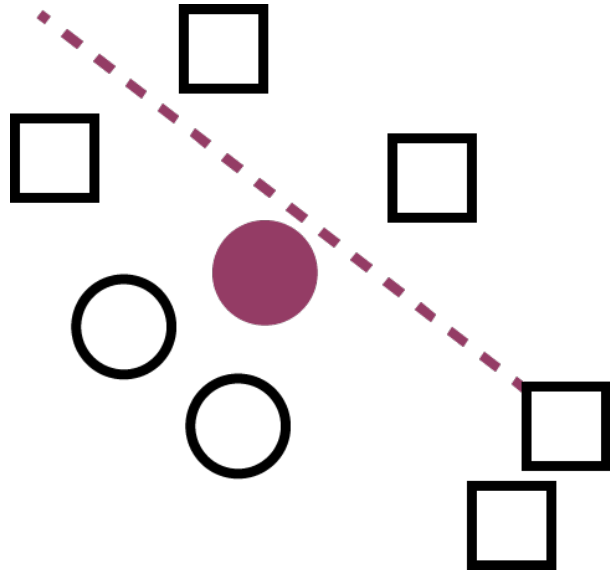
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Superiore*



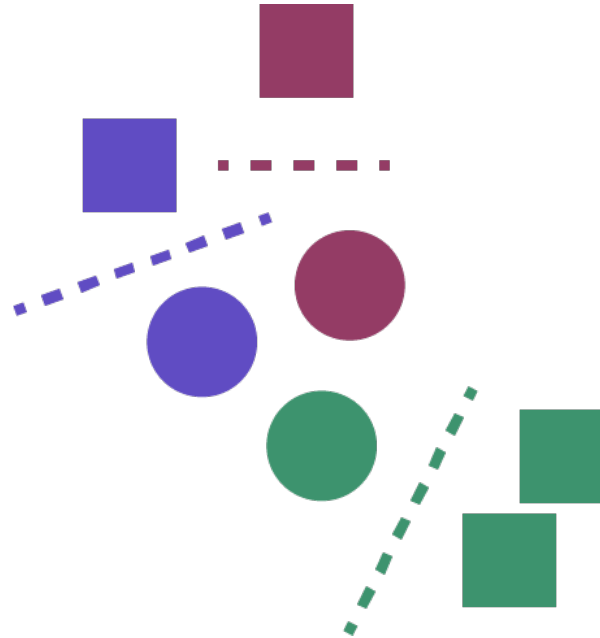
Explanation scope

Local



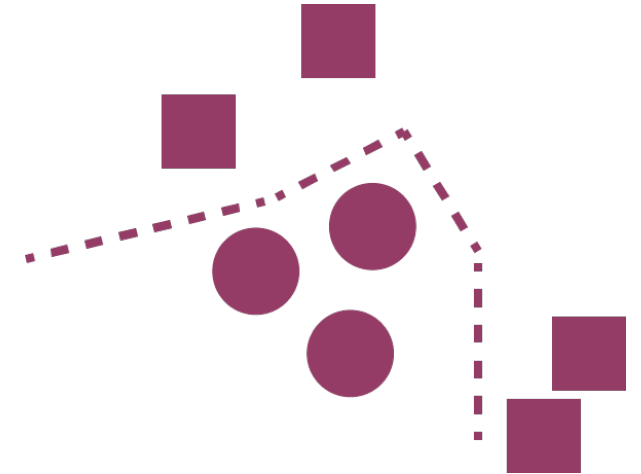
One instance, one
explaining model

Meso



One group, one explaining
model

Global



Many instances, one
explaining model

Meso-level: where are we?

Local-first

Explain locally, then merge explanations

- Off the shelf local explanation algorithms

- Limited explanation type
- Computationally expensive

Global-first

Partition data into groups, *then* explain

- Cohesive groups
- Meso explanations by design

- Optimized for accuracy
- Unclear trimming

Explainable AI for Trees: From Local Explanations to Global Understanding. Lundberg et al.

When Should We Use Linear Explanations? Delunay et al.

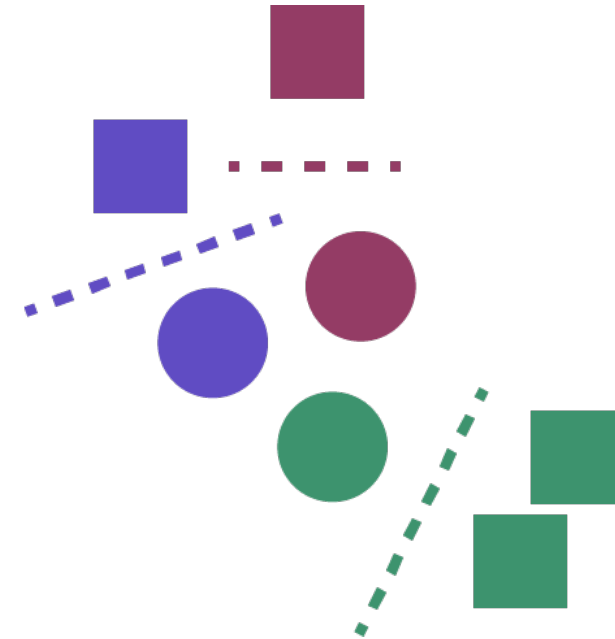
Interpretable Data Partitioning Through Tree-Based Clustering Methods. Guidotti et al.

Faithful and Customizable Explanations of Black Box Models. Lakkaraju et al.

Meso-level explanations: GrouX

We propose a framework for **Group**
e**X**plainability that provides:

- *Feature importance* explanations by design
- Meso-level scope
- Flexible grouping



Meso-level: the two objectives

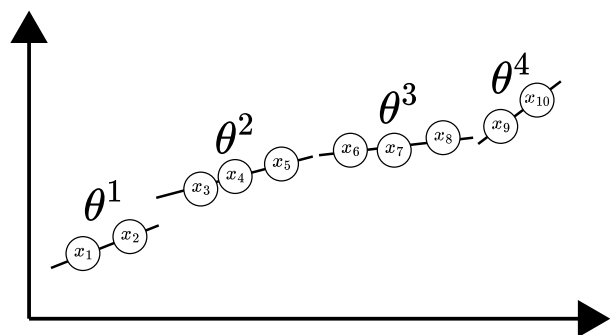
Discover groups which are

1. Cohesive and separate, i.e., clusters
2. Linearly separable

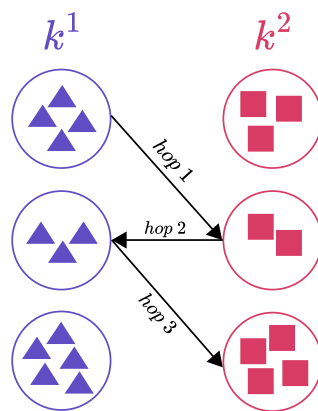
Jointly, this is a hard problem.

Key idea: encode inter-instance linear relationships in data, then look for cohesive groups!

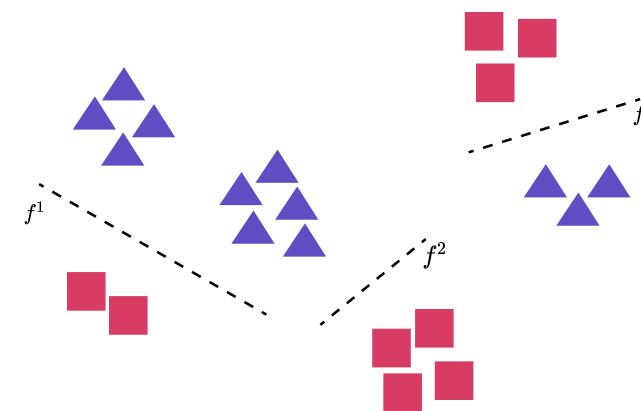
GrouX: the pipeline



1. Encode linear relationships Encode linear separability in the data.



2. Discover meso groups Cluster encoded instances into cohesive groups.



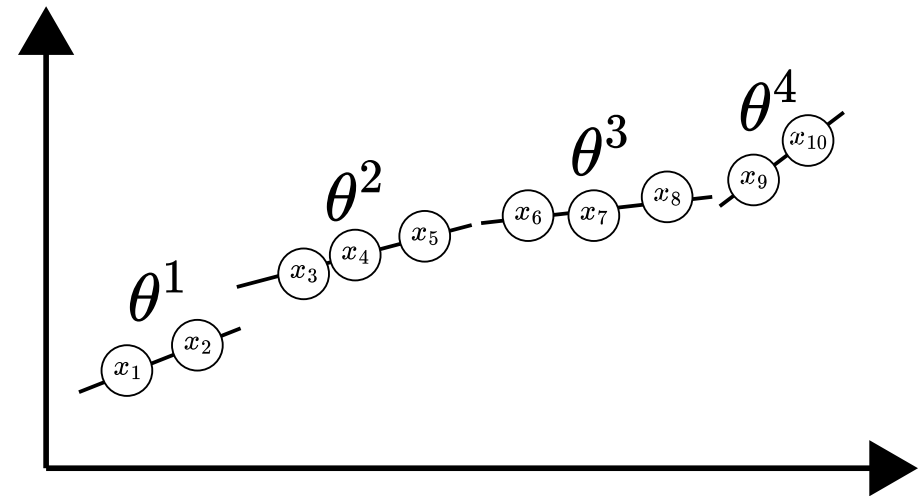
3. Learn meso models Learn a linear model on each group.

1. Learn linear relationships

How to encode linear separability? With linear approximation

Instances with similar values are approximated by the same linear model:

similar model parameters $\implies$ similar instances, and a shared linear relationship.

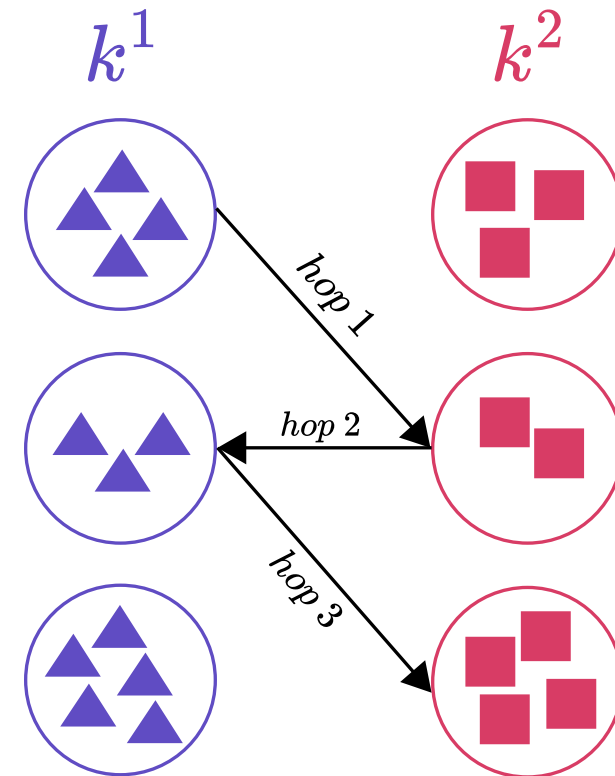


Piece-wise Linear Approximation (PLA): given a sequence of values, it approximates them with error-bounded linear splines.

2. Discovering meso groups

Classical clustering is label-oblivious:
clusters may be label-unbalanced, thus
resulting in poor explanations

GrouX leverages *fair clustering*
(Clusterlets), and discovers balanced
meso-level groups

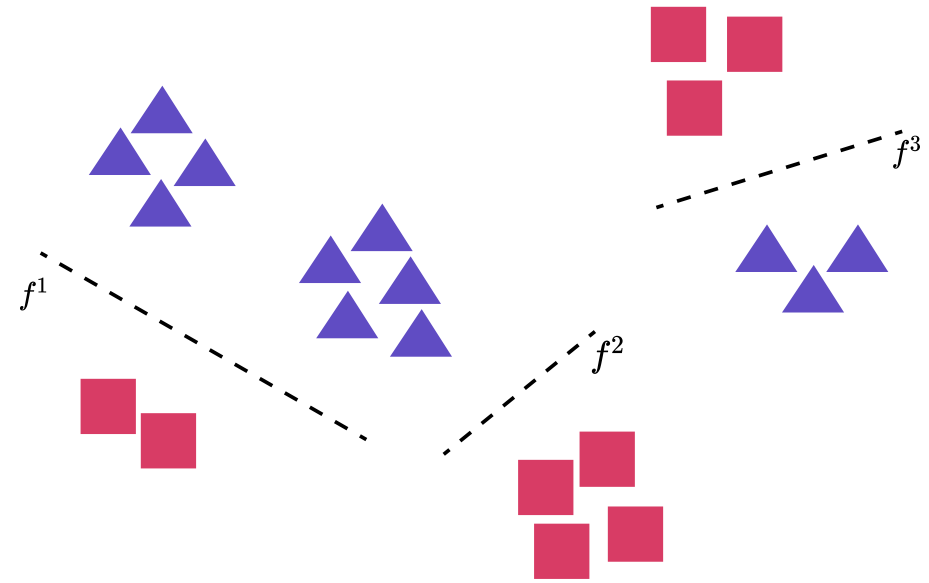


Creating clusters by merging clusterlets of classes k^1 (in blue) and k^2 (in red).

3. Learn meso-level models

Straightforward: learn a linear model, e.g., logistic regressor, on each meso model.

- **Explanation:** Linear model coefficients
- **Inference:** Closest model predicts



Meso-level linear models f^1, f^2, f^3 , learned on the respective meso-level group.

Research questions

Meso models

How accurate and complex are the meso-level explanations?

Inference

Can we use the meso-level models as standalone predictors?

Meso models

How accurate and complex are the meso-level explanations?

Model selection per dataset, performance and number of meso models (size).

Model	F1	Size
<i>k</i> -means	.703 $\pm$.139	7.0 $\pm$ 6.1
ParTree	.693 $\pm$.160	5.0 $\pm$ 3.6
FParTree	.762 $\pm$.178	21.1 $\pm$ 54.7
GrouX	.841 $\pm$.178	22.4 $\pm$ 54.7

Inference

Can meso models predict at global level?

GrouX VS Baseline VS Global interpretable models, model selection per dataset.

Model	F1	Size
<i>k</i> -means	.668 $\pm$.112	9.4 $\pm$ 4.9
ParTree	.663 $\pm$.086	4.2 $\pm$ 3.2
FParTree	.683 $\pm$.119	8.7 $\pm$ 6.4
GrouX	.738 $\pm$.112	16.2 $\pm$ 23.2
DT	.723 $\pm$.114	133 $\pm$ 149.0
LinearDT	.690 $\pm$.139	11.7 $\pm$ 3.71
Logistic	.676 $\pm$.135	

GrouX: Group Explainability through Local Approximation

Interpretable by design model

- Meso level
- Linear explanations
- More accurate than competitors, and on par with global models
- Controllable complexity
- github.com/msetzu/groux



Explainability scope: a continuum

Local-first

Local averaging

Average local
explanations

*Local Greedy
building*

Accumulate
instance in a meso
group, then explain
it

Global-first

Meso-models

Partition instances
into meso groups
with explanations

*Local to global
models*

Generalize local
explanations into
global ones: local to
global!

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Experimental setting

Competitors

- Baseline: k-means
- Meso-level:
ParTree, FParTree
- Inference:
Decision Tree,
Linear Tree,
Logistic

Datasets

- 12 binary
classification
datasets
- Strong class
imbalance:
 $26\% \pm 11\%$
minority class

Ablation

- Number of
clusterlets
- Linear encoding
- Clusterlet
matchers